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Development of a Wearable fNIRS System for Non-invasive Cerebral Oxygenation Monitoring

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Abstract. Near-infrared spectroscopy (NIRS) is a non-invasive technique extensively employed to monitor real-time data regarding tissue oxygenation levels, hemodynamics, and metabolic activity. By utilizing infrared light to penetrate biological tissue and analyzing the absorption and scattering of light, NIRS offers critical insights into oxygen saturation and blood flow dynamics. This paper details the design and development of a cost-effective, wearable, multichannel functional NIRS (fNIRS) device for measuring regional cerebral oxygen saturation (ΔrSO_2) across the prefrontal (fp1, fp2, f7, f8) and frontal (f3, f4) cortex regions. The device integrates optodes with four infrared wavelengths (660, 750, 850, and 880 nm) mounted on a flexible printed circuit board (PCB), ensuring optimal orthogonal contact with the skin and enhancing light penetration for improved signal strength. To validate the device's performance, a forearm occlusion experiment was conducted, and the results were compared against a standard cerebral oximeter device. Results indicated a Mean Absolute Error of 0.83% in ΔrSO_2 and a Mean Percentage Error of 0.89%. The findings confirm the system's capability for accurate, real-time cerebral oxygenation monitoring, highlighting its potential for clinical and research applications where affordability, portability, and reliability are essential.

Keywords: fNIRS, Cerebral Oxygen Monitoring, Data Acquisition, Flexible Printed Circuit Board

1 Introduction

Monitoring the hemodynamics of the brain is a crucial element of neuroimaging. The study of the brain's hemodynamic response becomes important as it can play a major role in the early detection of neurological disorders such as brain injury and stroke. Functional near-infrared spectroscopy (fNIRS) is increasingly favored as a tool for cross-cultural neuroimaging research [1]. Established neuroimaging modalities,

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including functional magnetic resonance imaging (fMRI) and positron emission tomography (PET), have been extensively employed to visualize cerebral functioning in humans [2] alongside Computer Tomography (CT) scans. These techniques are regarded as the gold standard in medical therapeutics for detecting hemodynamic disorders in clinical settings [3]. Each of these techniques has its advantages and limitations. Functional magnetic resonance imaging (fMRI) has substantially advanced the understanding of the brain's functional architecture [4]. However, it is limited by its lack of portability, low temporal resolution, and compatibility issues [4]. Likewise, there are limitations with PET as it necessitates restricted movement, confinement, and the administration of radioactive substances [2]. The limitations mentioned above necessitate the development of techniques to address and overcome them.

Near-infrared spectroscopy (NIRS) is a technique capable of addressing numerous challenges associated with other imaging modalities. It was introduced in 1977 by F. F. Jobsis as a non-invasive method for monitoring oxygen sufficiency [5]. Functional Near Infrared Spectroscopy (fNIRS) is a non-invasive optical imaging technique utilized for monitoring and recording of hemodynamic activity in the brain's cortical region [6]. fNIRS can be used to measure changes in concentrations of oxyhemoglobin (ΔHbO_2) and deoxyhemoglobin (ΔHbO) in the blood. This technique involves the use of infrared LEDs of varying wavelengths to emit light, while photodiodes collect the reflected light. Near Infrared Light (NIR) is directed onto the scalp, penetrates the skull, and subsequently disperses across brain tissues due to photon scattering [7]. NIR penetrating the tissue up to a few centimeters, interacts with blood chromophores, and the photodiode collects the reflected light [3]. Based on the difference between emitted light intensity and reflected light intensity, the concentration of oxyhemoglobin (HbO_2) and deoxyhemoglobin (HbO) can be calculated using the modified Beer-Lambert Law.

The advantages of fNIRS over fMRI include its non-invasive nature, lower cost, portability, superior temporal resolution, and reduced sensitivity to motion artifacts [8]. fNIRS devices can be wearable and wireless, making them useful for real-time applications. Early signs of a brain stroke can be detected using fNIRS, as indicated by a decrease in oxyhemoglobin concentration in and around the affected region [9]. Owing to its many advantages, fNIRS can complement clinical interviews and mental state assessments in the context of psychiatric diagnosis and treatment monitoring [10]. Additionally, the study of stress and decision-making can be augmented through the application of neuroimaging techniques such as electroencephalography (EEG) and fNIRS [11].

This paper presents the design and development of a low-cost, wearable multi-channel fNIRS device to monitor the oxygenation level of frontal and pre-frontal cortex areas. The innovation of this device is attributed to the strategic positioning of optodes at specific sites (fp1, fp2, f3, f4, f7, and f8) to facilitate localized measurement. At wavelengths of 850 nm and 880 nm, oxyhemoglobin (HbO_2) serves as the principal absorber, whereas at 750 nm and 660 nm, deoxyhemoglobin (Hb) is the predominant absorber [3]. For each region, a dedicated optode placement enhances spatial resolution for localization of low regional cerebral oxygenation saturation (rSO_2) areas, and the utilization of four wavelengths of light enhances the device's ability to capture variations in oxyhemoglobin and deoxyhemoglobin. To assess the accuracy of the developed

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device, it was benchmarked against a standard clinical device used for measuring regional cerebral oxygen saturation. Additionally, its functionality was evaluated through a forearm compression–decompression protocol conducted on five participants. The results demonstrated a consistent and reliable measurement of oxygen saturation, confirming the device’s suitability for non-invasive hemodynamic monitoring.

2 Optical Background

fNIRS can measure changes in ΔrSO_2 with the help of oxyhemoglobin (ΔHbO_2) and deoxyhemoglobin (ΔHbO). Based on the fundamental facts that human tissue is relatively transparent to light in the NIR window (650-1000 nm), NIR light is either absorbed by the chromophores or scattered in tissues, with scattering being more likely than absorption [12]. Near-infrared light sources are employed to radiate light. The emitted light experiences three phenomena: absorption, scattering, and reflection. The principal light absorbers are the chromophores. To maximize light absorption by oxyhemoglobin and deoxyhemoglobin, the wavelength of the produced light is meticulously chosen by examining the molar extinction coefficients of oxyhemoglobin and deoxyhemoglobin at various wavelengths. Fig. 1 shows the plot of molar extinction coefficient versus wavelength.

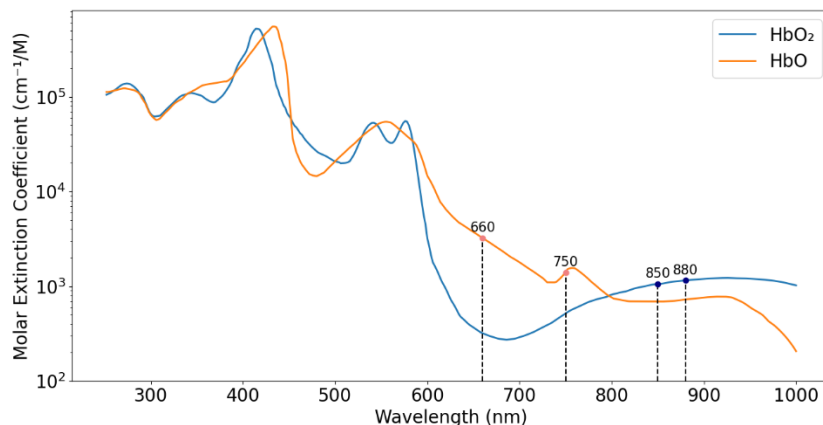


Fig. 1. Molar Extinction Coefficient versus Wavelength

fNIRS utilizes the Beer-Lambert Law to determine the change in the concentration of oxyhemoglobin and deoxyhemoglobin. Infrared light is emitted on the area of interest, and reflected light is collected by the detector. The difference between the intensity emitted and intensity collected back, we can estimate relative changes in the concentration using parameters like molar absorption coefficient (α), attenuation of light (A) and mean path length of detected photon (L) [4]. Using the parameters as mentioned above, the change in oxyhemoglobin concentration [ΔHbO_2] and the change in deoxyhemoglobin [ΔHbO], can be calculated using Eq. 1 and Eq. 2 [4]. Using ΔHbO_2 and

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ΔHbO_2 , ΔrSO_2 can be calculated using Eq. 4 and Eq. 3. ΔHbt denotes the total change in hemoglobin concentration.

$$\Delta\text{HbO}_2 = \frac{\alpha_{\text{Hb}}^{\lambda_1} \frac{\Delta A^{\lambda_2}}{L^{\lambda_2}} - \alpha_{\text{Hb}}^{\lambda_2} \frac{\Delta A^{\lambda_1}}{L^{\lambda_1}}}{\alpha_{\text{Hb}}^{\lambda_1} \alpha_{\text{HbO}_2}^{\lambda_2} - \alpha_{\text{Hb}}^{\lambda_2} \alpha_{\text{HbO}_2}^{\lambda_1}} \quad (1)$$

$$\Delta\text{Hb} = \frac{\alpha_{\text{HbO}_2}^{\lambda_1} \frac{\Delta A^{\lambda_2}}{L^{\lambda_2}} - \alpha_{\text{HbO}_2}^{\lambda_2} \frac{\Delta A^{\lambda_1}}{L^{\lambda_1}}}{\alpha_{\text{HbO}_2}^{\lambda_1} \alpha_{\text{Hb}}^{\lambda_2} - \alpha_{\text{Hb}}^{\lambda_1} \alpha_{\text{HbO}_2}^{\lambda_2}} \quad (2)$$

$$[\text{Hbt}] = [\text{Hb}] + [\text{HbO}_2] \quad (3)$$

$$\Delta\text{rSO}_2 = \frac{[\Delta\text{HbO}_2]}{[\Delta\text{Hbt}]} \times 100 \quad (4)$$

3 Methodology

The designed device is a continuous-wave (CW) near-infrared spectroscopy (NIRS) system that employs the modified Beer-Lambert law to calculate relative changes in hemoglobin concentrations. The choice of a CW configuration with non-switching LEDs is justified by its simplicity, cost-effectiveness, and suitability for real-time monitoring of hemodynamic responses [13]. The device has three main blocks to monitor the hemodynamics of the area of interest of the brain. The fNIRS probe, the amplifier, and the microcontroller. Together, these three blocks will sense, amplify, process, and extract the bio-markers.

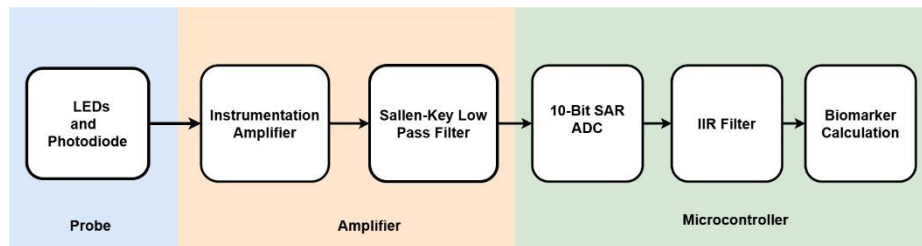


Fig. 2. Block Diagram of the system comprises of three main stages namely: Probe, Amplifier, and Microcontroller

3.1 fNIRS Probe

The probe is fixed in a skull cap and is directly placed on the head. It is made to cover the pre-frontal and some parts of the frontal cortex, namely the fp1, fp2, f3, f4, f7, and f8 regions of the brain, and the optodes are placed following the 10/20 system. The

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probe has twenty-four infrared LEDs to emit light on the mentioned region, and six photo-diodes capture the reflected light. Six photo-diodes are dedicatedly placed for each location (fp1, fp2, f3, f4, f7, and f8) that collects light in the wavelength range of 430 nm to 1100 nm. The four IR LEDs used are of four different wavelengths, 660 nm (SML-LX1206SRC-TR), 750 nm (OIS-330-740-X-T), 850 nm (15414185BA210), and 880 nm (KP-2012SF4C). A set of four IR LEDs (660 nm, 750 nm, 850 nm, and 880 nm) and a photo-diode are placed for each location (fp1, fp2, f3, f4, f7, and f8). These elements are placed in a square-shaped arrangement, where the IR LEDs occupying the corners and the photo-diode holds the center of the square as shown in Fig. 3. LED A (880 nm) and LED C (850 nm) are placed on the same diagonal and LED B (750 nm) and LED D (660 nm) are placed on the other diagonal of the square. This arrangement ensures proper coupling of LEDs with the photodiode. Two LED sources with distinct wavelengths (850 and 880 nm) exhibit greater absorption by oxyhemoglobin (HbO₂), whereas two more LED sources with wavelengths (660 and 750 nm) show increased absorption by deoxyhemoglobin (Hb) [3].

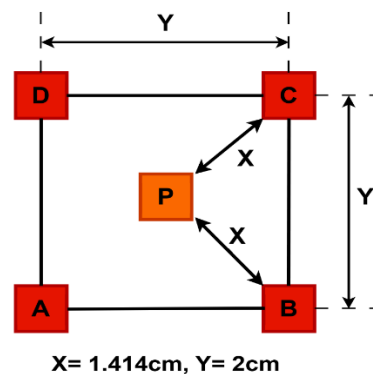


Fig. 3. Arrangement of IR LEDs and Photodiode

All the components are mounted on a flexible PCB to keep them in place and adapt to the forehead's curved surface. This allows photodiodes to maintain orthogonal contact with the skin and prevent direct photon leakage, allowing more light to go inside the skin. This also improves the signal strength of the reflected signal. In a single optode, the distance between an LED and a photodiode is 1.414 cm, and the distance between two adjacent LEDs is 2 cm. Fig. 3 shows the arrangement of IR LEDs and photodiodes in a single optode, and Fig. 4 depicts the arrangement of all six optodes in accordance with the 10/20 system to cover fp1, fp2, f3, f4, f7, and f8 regions of the brain.

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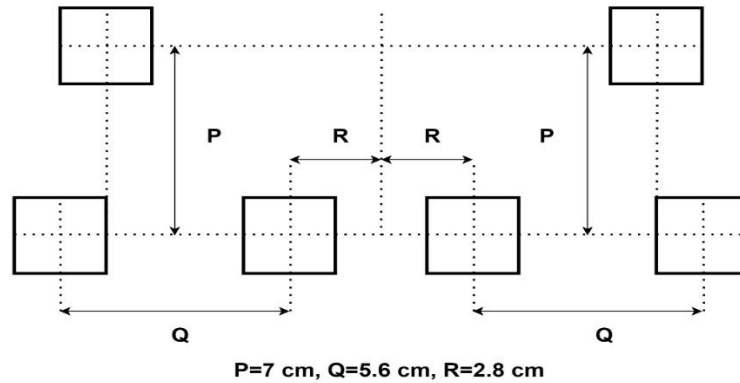


Fig. 4. Arrangement of Optodes

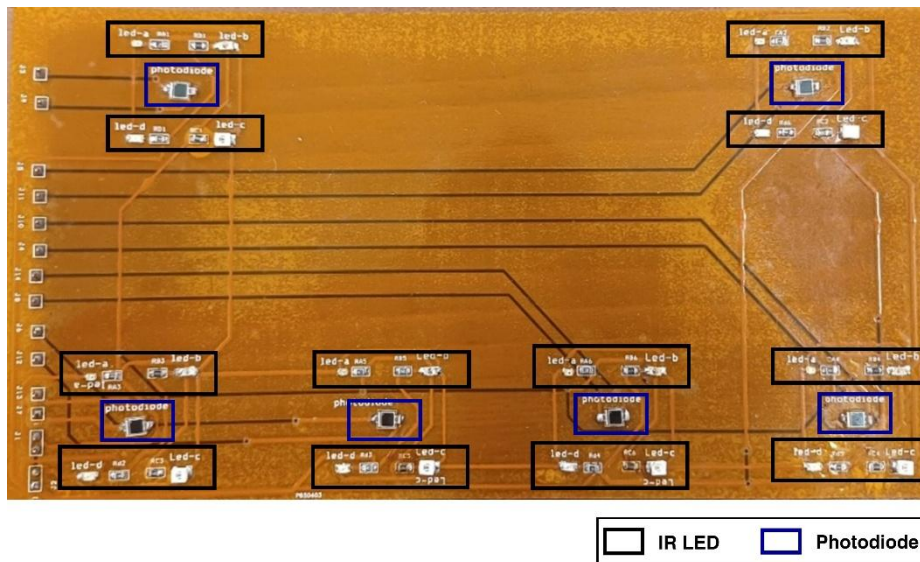


Fig. 5. Flexible PCB with IR LEDs and Photodiode

3.2 Amplifier Circuit

The Amplifier circuit is employed to amplify the weak analog signal from the photodiode. On careful observation, the photodiode signal is found to range between 50mV to 100mV and hence needs amplification. Weak signals with low working frequency can easily get affected by noise [14]. The noise can originate from the circuit, components, high-frequency signal, or power line noise [15]. An instrumentation amplifier with a gain of 30dB is chosen to address the above-mentioned problems and suppress noise as it has good stability with a high Common-mode rejection ratio (CMRR) and high input impedance [16]. To further eliminate the effect of power line interference and the high-

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frequency noise from the signal, an active 2nd order, unity gain Sallen and Key low pass filter with a cutoff frequency of 30Hz is employed. LM358P general-purpose op-amp IC is used to implement both an instrumentation amplifier and a Sallen and Key low-pass filter. The signal is then fed to the microcontroller unit for further processing. Fig. 6 shows the photodiode bias circuit as well as the instrumentation amplifier. Fig. 7 shows the Sallen and Key low-pass filter circuit.

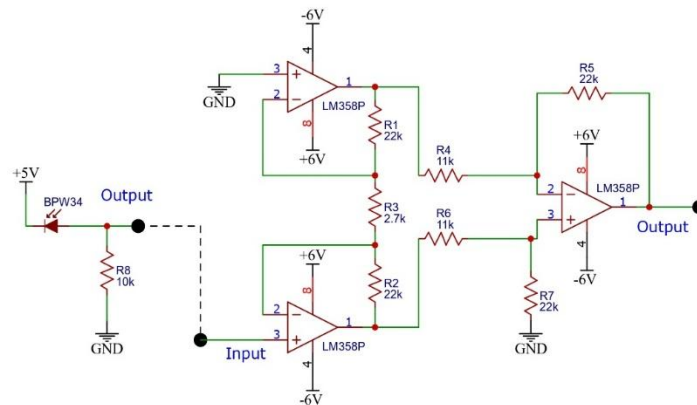


Fig. 6. Instrumentation Amplifier Circuit

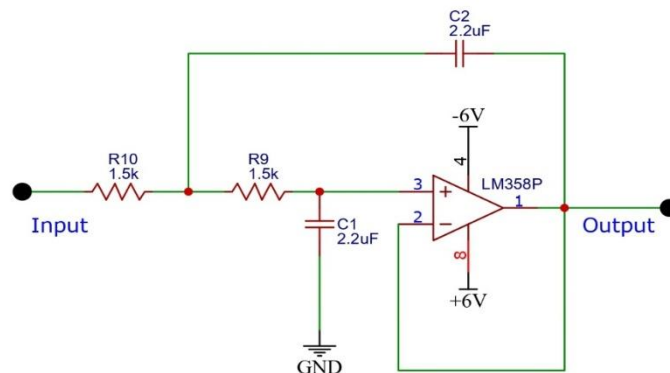


Fig. 7. Sallen and Key Low-pass filter

3.3 Microcontroller Unit

The microcontroller unit converts the amplified analog signal to digital signal. For the conversion, a 10-bit SAR (successive approximation register) type ADC known for a good balance between speed and accuracy is used. Afterwards, the digital signal is fed to the microcontroller ATmega 2560, where the signal is filtered out by an IIR filter of 30 Hz to further smooth out the signal. After the filtering process and removing the high-frequency artifacts, the signal is now ready for the extraction of the bio-markers.

4 Experiment Protocol

Previous research conducted this artery occlusion experiment to evaluate their technology [3], [4], [17]; hence, we have replicated the experiment to assess the performance of our device. A total of five healthy male participants (mean age: 24.71 years; standard deviation: 1.03 years) were recruited for the study at the Indian Institute of Technology Mandi, India. Eligibility criteria included being in good health and having no physical obstructions or injuries in the forearm region that could interfere with data acquisition. The study protocol was reviewed and approved by the Institutional Scientific Committee on Human Research and the Ethical Review Board (Reference No: IIT M/IEC(H)/2022/SRC/P).

In this experimental setup, as shown in Fig. 8, a flexible near-infrared spectroscopy (NIRS) probe is affixed to the subject's forearm, approximately 7 - 8 cm proximal to the wrist, ensuring close contact with the skin to maintain signal fidelity. Positioned 3 - 5 cm above the NIRS probe is a standard cerebral oximeter device called VO200 - NeurOs Cerebral Oximetry (Mespere LifeSciences, Waterloo, Canada) facilitating concurrent monitoring of regional oxygen saturation (ΔrSO_2) in both cerebral and peripheral tissues. Additionally, a pressure cuff is secured around the upper arm, one inch above the elbow crease. The subject remains seated with the forearm resting on a table throughout the procedure. The experiment spans three minutes, during which data is sampled at an approximate sampling rate of 16 Hz. The sampling rate is limited by the simultaneous recording from four different photodiodes, processing the data in real-time on Arduino and transmitting to the computer for recording. In this way, continuous monitoring of ΔrSO_2 is conducted to assess tissue oxygenation dynamics under varying physiological conditions.

Phase 1 (0 - 1 minute) Baseline Measurement: During the initial minute, the subject is at rest, establishing a baseline for ΔrSO_2 levels. This phase serves as a control to compare subsequent changes induced by vascular occlusion and reperfusion.

Phase 2 (1 - 2 minutes): Induced Ischemia and Recovery: In the first 30 seconds of the second minute, the pressure cuff gradually inflated, reducing arterial inflow to the forearm and inducing a state of transient ischemia. This maneuver results in a measurable decline in ΔrSO_2 , reflecting diminished tissue oxygenation. Following this, the cuff is slowly deflated over the next 30 seconds, initiating reperfusion. During this recovery phase, ΔrSO_2 levels progressively rise, approaching baseline values within approximately 45 to 50 seconds, indicating restoration of normal blood flow and oxygen delivery.

Phase 3 (2 - 3 minutes): Stabilization: The final minute allows for the stabilization of ΔrSO_2 levels, confirming the re-establishment of baseline tissue oxygenation and ensuring the reversibility of the induced ischemic event.

This protocol effectively demonstrates the responsiveness of NIRS technology in detecting rapid changes in tissue oxygenation, providing valuable insights into vascular and metabolic function under controlled conditions.

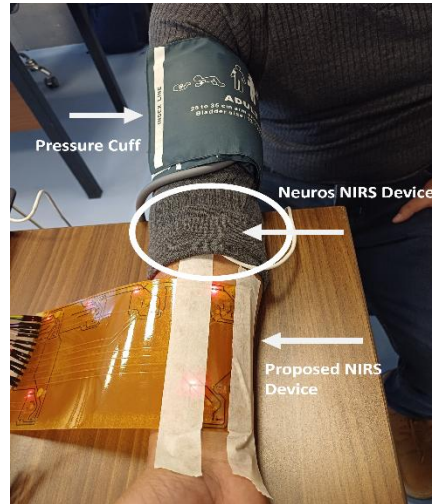


Fig. 8. Complete Experiment Setup showing

5 Result

5.1 Comparison with the standard device

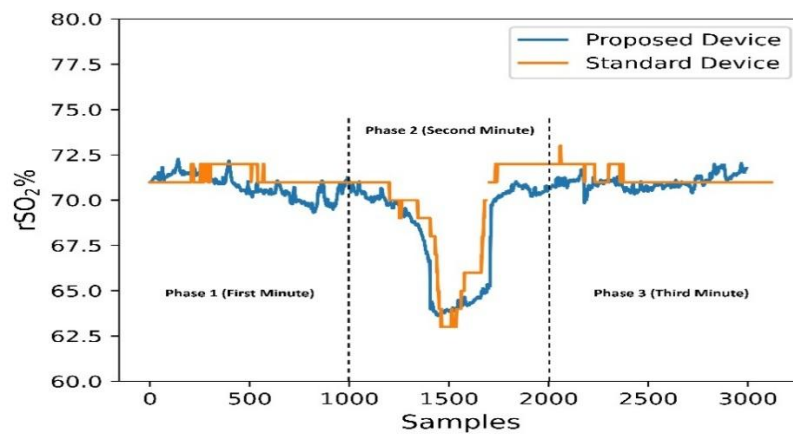


Fig. 9. ΔrSO_2 Variation During Experiment. Comparison of developed device (orange line) with VO200 - Neuros Cerebral.

The experimental results confirm that the gadget operates well and aligns with the output of the standard device. Fig. 9 shows the complete variation of the ΔrSO_2 level throughout the experiment. During the first minute of observation, both devices confirm that the baseline reading of ΔrSO_2 fluctuates between 70% and 71%. During the first half of the second minute, the pressure cuff begins to apply pressure to the upper arm,

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thereby diminishing blood flow. The ΔrSO_2 level initially declines gradually, followed by a dramatic decrease to 64 % when blood flow exceeds a minimum threshold. The standard device exhibits a rapid decline in the ΔrSO_2 level, but our device demonstrates a progressive decrease in ΔrSO_2 level corresponding to the applied pressure.

This indicates that our gadget has good sensitivity for ΔrSO_2 levels. In the latter part, the pressure begins to diminish gradually, resulting in an incremental increase in blood flow to the arm. As blood flow increases, the ΔrSO_2 level begins to rise. Both devices successfully detected this upward trend; however, our device exhibited a steady increase in the ΔrSO_2 level, indicating superior sensitivity, while the standard device demonstrated a sudden spike in the ΔrSO_2 level.

At the three-minute mark, blood flow returns to baseline, and the ΔrSO_2 level stabilizes within the 70-71% range, as validated by both devices.

5.2 Comparison among the participants

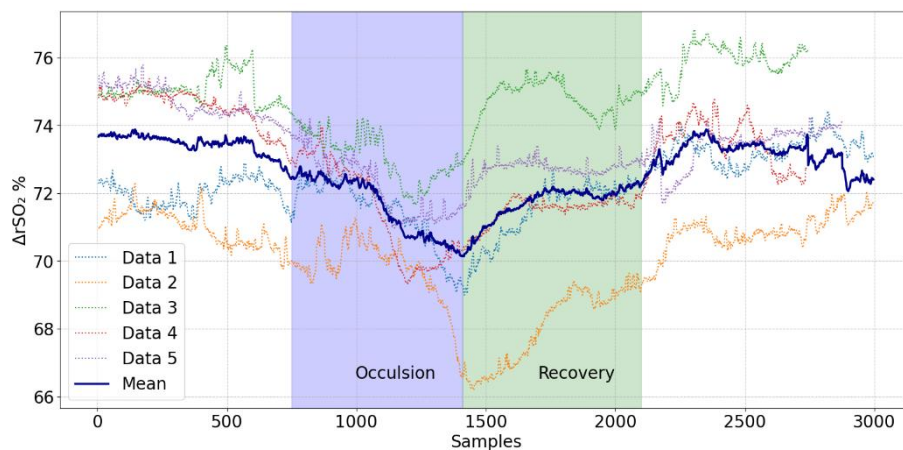


Fig. 10. ΔrSO_2 responses from five participants during occlusion (blue) and recovery (green) phases. The mean trend (dark blue) shows a consistent decrease in oxygen saturation during compression and recovery post-release.

Fig. 10, illustrates the ΔrSO_2 waveforms recorded from five participants during the forearm occlusion and recovery phases. Each colored trace corresponds to an individual participant's response, while the dark blue curve represents the mean ΔrSO_2 response across all participants. The shaded blue region represents the occlusion period, during which a gradually increasing external pressure was applied via a cuff, leading to a reduction in regional blood oxygen saturation. The green region denotes the recovery phase, where the applied pressure was systematically released, allowing for reperfusion of the tissue.

A consistent pattern is observed across subjects: oxygen saturation levels decline during the occlusion phase and progressively return toward baseline during recovery. The averaged trend line clearly reinforces this pattern, confirming the device's capability to

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detect physiologically relevant changes in tissue oxygenation in response to vascular occlusion and reperfusion.

6 Conclusion

The findings of this study demonstrate that the developed low-cost, wearable, and multichannel fNIRS system provides accurate monitoring of regional cerebral oxygen saturation (ΔrSO_2) across the prefrontal and frontal cortex regions. Performance validation through a forearm occlusion experiment, benchmarked against a standard device used for measuring cerebral oxygenation, yielded a Mean Absolute Error of 0.83% and a Mean Percentage Error of 0.89%, thereby confirming the system's precision and reliability. The enhanced sensitivity of the device was evident in its capacity to detect gradual changes in ΔrSO_2 during both the occlusion and recovery phases, exhibiting comparable performance to the standard device. Consistent trends were observed across all five participants during pressure application and recovery phases, further supporting the robustness of the system. The integration of flexible optodes, a compact form factor, and the use of low-cost components enhances the device's suitability for both clinical and research contexts, particularly in resource-constrained settings. Overall, the study underscores the potential of the proposed fNIRS system as a reliable, accessible, and effective solution for non-invasive cerebral oxygenation monitoring.

7 Acknowledgement

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